



Tensor Transformer for hyperspectral image classification

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ABSTRACT

Hyperspectral image (HSI) is widely used in real-world classification tasks since it contains rich spatial and spectral features consisting of hundreds of continuous bands. In recent years, the deep learning-based HSI classification methods, such as convolutional neural network (CNN) and Transformer, have achieved good performance in HSI classification tasks. Indeed, it is acknowledged that Transformer-based neural networks, owing to their remarkable capacity to extract long-range features, frequently outperform CNN-based neural networks in HSI classification scenarios. However, Transformer-based methods always require the sequentialization of the raw 3-D HSI data, potentially disrupting the spatial-spectral structural features. This shortcoming has degraded the classification accuracy of HSI data. In this paper, we proposed a Tensor Transformer (TT) framework for HSI classification. The TT model is an end-to-end network that directly takes the raw HSI tensor data as the input sample, without the need for raw data sequentialization. The core component of the proposed framework is the Tensor Self-Attention Mechanism (TSAM), which enables the network to efficiently extract long-range spatial-spectral structural features without losing the inherent structural relationships inner the sample. Through extensive experiments on four widely used HSI datasets, the proposed TT model demonstrates superior classification performance in discriminating land features with similar spectrum compared to state-of-the-art methods.

1. Introduction

Hyperspectral image (HSI) encompasses numerous contiguous spectral bands, comprising an abundance of spectral data [1,2]. Since ground features have different characteristics in different dimensions, the rich spectral dimensions create favorable conditions for precise ground feature classification [3,4]. The classification endeavors within hyperspectral imaging encompass a diverse array of applications, including ecological science [5], precision agriculture [6–8], forest monitoring [9], and climate prediction [10].

In previous studies on HSI classification, numerous conventional machine learning techniques were employed for the task, including support vector machine [11], random forest [12], and K-nearest neighbor [13]. Nevertheless, these approaches exhibit limited efficacy in extracting advanced-level features from HSIs, consequently resulting in diminished classification accuracy.

In recent times, HSI classification techniques grounded in deep learning have gained significant prominence. This surge in popularity stems from the capability of deep learning to effectively extract deep abstract features, gradually supplanting the preceding classification models. Deep learning yields substantial advancements in the performance of HSI classification. Guo et al. [14] introduced a methodology that integrated a pretrained stacked autoencoder (SAE) model

for dimensionality reduction into a CNN classifier. Hamida et al. [15] proposed a 3D-CNN-based framework for HSIC, where 3D convolutions were utilized to more effectively capture the spectral-spatial dependencies present in hyperspectral images, outperforming traditional 2D CNN networks. Mou et al. [16] employed recurrent neural networks (RNNs) for HSI classification and introduced an RNN classification network to mine the relationships among spectral bands. Song et al. [17] proposed MSLAN, a two-branch multidirectional spectral-spatial LSTM network for hyperspectral image classification, which captures dependencies in six directions using LSTM and incorporates a feature fusion module to improve classification performance, particularly with limited training samples. Although CNN-based methods excel at capturing local context information from HSIs, they often struggle to discern subtle spectral feature discrepancies among non-neighboring spectral bands, thereby limiting their effectiveness in this regard. RNN-based methods, due to their heavy reliance on the sequential order of spectral bands, encounter difficulties in learning long-range dependencies, resulting in a performance bottleneck for practical HSI classification.

Recently, Transformers [18] have showcased remarkable performance across a multitude of tasks in natural language processing (NLP) and computer vision (CV) domains. The Transformer architecture is

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founded upon the self-attention mechanism, which adeptly captures global features by discerning the relationships between elements within a sequence. Transformers and their variants have been effectively employed in numerous computer vision tasks, such as classification [19, 20], detection [21,22], and segmentation [23,24], yielding impressive results. This presents a novel and efficacious approach for HSI classification. With the rise of attention mechanisms, some attention-based modules were applied in HSIC, such as Residual spectral-spatial attention network (RSSAN) [25] and attention-based adaptive spectral-spatial kernel ResNet (A^2S^2K -ResNet) [26]. These approaches either integrate self-attention with convolutional networks or leverage self-attention to modify specific components of convolutional architectures, while preserving their general structure. However, the performance of these models in classification still heavily relies on the CNN backbone for feature extraction and representation. Compared to RGB images in the CV domain, HSI possess significantly higher spectral dimensionality, containing rich spectral information across multiple wavelengths. While CNNs are effective in extracting local spatial features, vision Transformers (e.g., ViT [27]) are particularly advantageous in capturing the long-range dependencies in the spectral dimension, leveraging their self-attention mechanism to model global spectral relationships more effectively. This makes Transformers well-suited for HSI classification tasks, where the spectral dependencies play a critical role in distinguishing different materials or land cover types. He et al. [28] presented a bi-directional encoder representation from Transformers, denoted as HSI-BERT, which endeavors to capture global dependencies. The proposed architecture exhibits notable generalization capabilities, enabling its extension to regions with varying shapes without necessitating retraining. Hong et al. [29] utilized the Transformer architecture as a substitute for CNNs in HSI classification, introducing a novel backbone network known as SpectralFormer. This network adeptly captures the sequence correlation information between distinct spectral segments of HSIs and is applicable for both pixel and patch input. A cross-layer jump connection mechanism was proposed that intelligently merges data from shallow layers to deep layers via adaptive learning. Yang et al. [30] proposed the HiT model, the Spectral-Adaptive 3D Convolution Projection (SACP) module utilizes two spectral-adaptive 3D convolution layers to generate local spatial-spectral information from hyperspectral images. Subsequently, depthwise convolution operations are utilized to encode this spatial-spectral information across the dimensions of height, width, and spectrum. Roy et al. [31] proposed morphFormer, a spectral-spatial attention network that integrates morphological convolution with Transformer models to capture both spatial and spectral features, enhancing hyperspectral image classification performance by effectively modeling complex dependencies and reducing computational complexity. Sun et al. [32] proposed the SSFTT model, which utilizes a hybrid approach combining convolutional and Transformer architectures for Spectral-Spatial feature extraction. Furthermore, it incorporates a Gaussian-weighted feature tokenizer to enhance semantic representation. Other similar works specifically designed for HSI classification can be found in literatures [33,34]. However, these methods generally require either the serialization or dimensionality reduction of HSI data, both of which may lead to the loss of structural features in the 3-D space, thereby limiting further improvements in HSI classification performance.

Based on the above-mentioned analysis, this paper aims to propose a novel Transformer-based HSI classification network, called Tensor Transformer (TT), to achieve the high-performance HSI classification by utilizing long-range spatial-spectral structural features. Specifically, TT consists of two key modules, namely, the Tensor Self-Attention Mechanism (TSAM) module and the Tensor Encoder Layer (TEL) module. Different from the existing Transformer-based HSI classification methods (such as [35]), our proposed TSAM module supports the direct input of tensor data without the need for sequence serialization. This feature distinguishes it from other models and is particularly advantageous in scenarios where input data naturally take the form

of tensors, which could capture long-range spatial-spectral structural features from the input HSIs. The responsibility of the TEL module is to encode the input tensor, extracting a sequence of feature representations from the initial data, which will serve the subsequent network's classification endeavor. In general, the main contributions of this work are as follows:

(1) The advantage of Transformers is their ability to extract long-range spectral-spatial features from HSI samples. However, the serialization of samples leads to the loss of structural features in the 3-D space, preventing Transformers from utilizing their strengths. The paper proposes dividing tensor samples along the spectral dimension as input to the Transformer, thus avoiding serialization and preserving spatial structural information. This allows the Transformer to effectively leverage its ability to extract long-range spectral-spatial features.

(2) We propose a TSAM module, which performs linear transformations from multiple dimensions of the tensor samples to extract high dimension structural features within the samples. Moreover, the input sub-tensor of TSAM module is derived from the segmentation of tensor sample along the spectral dimension, thus each of the traditional large size attention parameter matrices W_q , W_k , W_v corresponding to spectral length are replaced by three small factor matrices. This enables to significantly reduce the number of parameters in the Transformer.

(3) We provide a detailed analysis of the evaluation metrics, the number of parameters, FLOPs, and prediction time for TT in comparison to other methods on four benchmark datasets. The results demonstrate that the proposed TT outperforms state-of-the-art Transformer and CNN-based methods.

The remainder of this article is organized as follows. The second part presents the Tensor Self-Attention Mechanism (TSAM) module, Tensor Encoder Layer (TEL) module and the architecture of Tensor Transformer (TT). The third part provides experimental results to verify the feasibility of the proposed method. The fourth part discusses the experimental results. Finally, our conclusion is drawn in Section 5.

2. Proposed methodology

2.1. Tensor self-attention mechanism

HSI typically manifests as a three-dimensional tensor, wherein the initial two dimensions signify the size of the pixel's neighborhood window, while the third dimension corresponds to the spectral dimension. While traditional CNNs can directly accommodate three-dimensional tensor inputs, Transformer-based neural networks, on the other hand, are primarily designed to handle sequential data inputs. Therefore, when employing a Transformer-based neural network for hyperspectral data classification, it is often imperative to serialize the original data. Nonetheless, the serialization operation unavoidably destroys the inherent spatial or spectral features contained within the HSI. Consequently, this paper proposes a Tensor Self-Attention Mechanism (TSAM) which can perceive the long-range spatial-spectral structural features within the original data and extract features deeply without destroying the internal structural information of the HSI.

For each training sample tensor $\mathcal{X} = \{x_{i_1 i_2 i_3}\} \in \mathbb{R}^{I_1 \times I_2 \times I_3}$ of hyperspectral data, $I_1 \times I_2$ represents the spatial size of the pixel's neighborhood, and I_3 represents the number of spectral bands. The integration of the TSAM module in processing HSI has unleashed a vast array of possibilities for Transformer-based networks. Let us envision a scenario where the third dimension of a single sample is J . After overlapping sampling along the third dimension of $\mathcal{X}$ with a spectral band length of J , an input tensor group $\mathcal{Y}_1 \dots \mathcal{Y}_N$ containing N tensors can be obtained, where $\mathcal{Y}_i = \{y_{i_1 i_2 j}\} \in \mathbb{R}^{I_1 \times I_2 \times J}$. In the conventional computation of the self-attention mechanism, it is essential to create three weight matrices, namely W_q , W_k and W_v . These matrices are multiplied with the original input sequence to generate three matrices Q , K and V , which encapsulate the output of each sequence respectively.

In the TSAM module, the Tucker Transformation is employed to compute the output tensor based on the input tensor. Consequently, three sets of weight matrices are required to be constructed: $\mathbf{M}_1^q$, $\mathbf{M}_2^q$ and $\mathbf{M}_3^q$ for constructing the output tensor group $\mathcal{Q}_1 \dots \mathcal{Q}_N$. $\mathbf{M}_1^k$, $\mathbf{M}_2^k$ and $\mathbf{M}_3^k$ for constructing the output tensor group $\mathcal{K}_1 \dots \mathcal{K}_N$. $\mathbf{M}_1^v$, $\mathbf{M}_2^v$ and $\mathbf{M}_3^v$ for constructing the output tensor group $\mathcal{V}_1 \dots \mathcal{V}_N$. Among them, $\mathcal{Q}_i = \{q_{i_1 i_2 j}\} \in \mathbb{R}^{I_1 \times I_2 \times J}$, $\mathcal{K}_i = \{k_{i_1 i_2 j}\} \in \mathbb{R}^{I_1 \times I_2 \times J}$ and $\mathcal{V}_i = \{v_{i_1 i_2 j}\} \in \mathbb{R}^{I_1 \times I_2 \times J}$ respectively. Taking the construction of the output tensor $\mathcal{Q}_i$ as an example, the output tensor $\mathcal{Q}_i$ is acquired by executing the following tensor transformation on the input tensor $\mathcal{Y}_i$:

$$\mathcal{Q}_i = \mathcal{Y}_i \times_1 \mathbf{M}_1^q \times_2 \mathbf{M}_2^q \times_3 \mathbf{M}_3^q \quad (1)$$

where $\mathbf{M}_1^q \in \mathbb{R}^{I_1 \times I_1}$, $\mathbf{M}_2^q \in \mathbb{R}^{I_2 \times I_2}$ and $\mathbf{M}_3^q \in \mathbb{R}^{J \times J}$ is the factor matrix. Symbol $\times_1$ represents the mode product of $\mathbf{M}_1^q$ along the first dimension of $\mathcal{Y}_i$, symbol $\times_2$ represents the mode product of $\mathbf{M}_2^q$ along the second dimension of $\mathcal{Y}_i$, and symbol $\times_3$ represents the mode product of $\mathbf{M}_3^q$ along the third dimension of $\mathcal{Y}_i$. It is worth noting that the order of calculation for the mode product has no bearing on the final output tensor $\mathcal{Q}_i$. Each element $q_{i_1 i_2 j}$ in the output tensor $\mathcal{Q}_i$ can be computed using the following equation:

$$q_{i_1 i_2 j} = \sum_{i_1=1}^{I_1} \sum_{i_2=1}^{I_2} \sum_{j=1}^J y_{i_1 i_2 j} m_{1 i_2 p_2}^q m_{2 i_2 p_2}^q m_{3 j p_3}^q \quad (2)$$

where $m_{1 i_1 p_1}^q$, $m_{2 i_2 p_2}^q$ and $m_{3 j p_3}^q$ represent the elements in the transformation matrices $\mathbf{M}_1^q$, $\mathbf{M}_2^q$ and $\mathbf{M}_3^q$ respectively. By executing the aforementioned operation on each input tensor, the final output tensor group $\mathcal{Q}_1 \dots \mathcal{Q}_N$ can be derived. The calculation method of the output tensor groups $\mathcal{K}_1 \dots \mathcal{K}_N$ and $\mathcal{V}_1 \dots \mathcal{V}_N$ is similar to $\mathcal{Q}_1 \dots \mathcal{Q}_N$.

Upon acquiring the output tensor groups corresponding to the input tensor group, the TSAM module employs tensor dot product to compute the correlation between different output tensor groups. For the output tensor group $\mathcal{Q}_1 \dots \mathcal{Q}_N$, each element is dot producted with each element of the output tensor group $\mathcal{K}_1 \dots \mathcal{K}_N$ to obtain a coefficient matrix $U = \{\alpha_{nc}\} \in \mathbb{R}^{N \times N}$. The equation for computing each element in the coefficient matrix U is as follows:

$$\alpha_{nc} = \langle \mathcal{Q}_n, \mathcal{K}_c \rangle = \sum_{i_1=1}^{I_1} \sum_{i_2=1}^{I_2} \sum_{j=1}^J q_{i_1 i_2 j} k_{i_1 i_2 j} \quad (3)$$

where the symbol $\langle \rangle$ in Eq. (3) represents the inner product of two tensors, $\mathcal{Q}_n$ represents the n th element in the output tensor group $\mathcal{Q}_1 \dots \mathcal{Q}_N$, while $\mathcal{K}_c$ refers to the c th element in the output tensor group $\mathcal{K}_1 \dots \mathcal{K}_N$. The correlation coefficient matrix U is subsequently processed through a softmax operation, resulting in the generation of the correlation coefficient matrix U' . Each individual column vector within the derived correlation coefficient matrix U' is indeed sequentially multiplied element-wise with the corresponding elements of the output tensor group $\mathcal{V}_1 \dots \mathcal{V}_N$. This particular operation results in the overall output $\mathcal{O}_1 \dots \mathcal{O}_N$ of the TSAM module. Taking the initial element $\mathcal{O}_1$ and $\mathcal{O}_n$ from the ultimate output tensor group $\mathcal{O}_1 \dots \mathcal{O}_N$ as an illustrative example, the comprehensive operation process of the tensor self-attention mechanism is depicted in Fig. 1. The operation process for the remaining elements follows a similar pattern.

2.2. Tensor encoder layer

Based on the proposed tensor self-attention mechanism in this chapter, we propose a Tensor Encoder Layer (TEL) module. The architecture of the TEL module is shown in Fig. 2.

As illustrated in Fig. 2, after applying layer normalization to the input tensor, the TSAM module is utilized to compute the output. Subsequently, the summation result also undergoes layer normalization. Finally, the normalized summation result is added to the summation result, serving as the output of the TEL module. It is noteworthy to mention that, akin to the Encoder layer in the conventional Transformer model, the input data size and output data size of the TEL

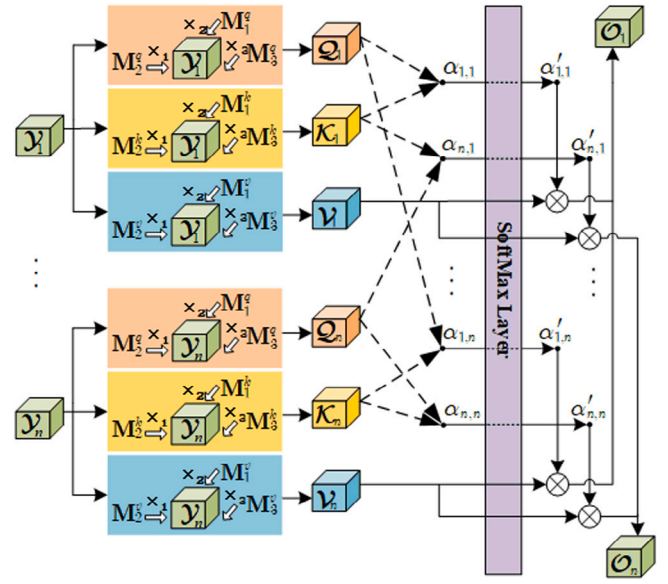


Fig. 1. The operation process of the TSAM module.

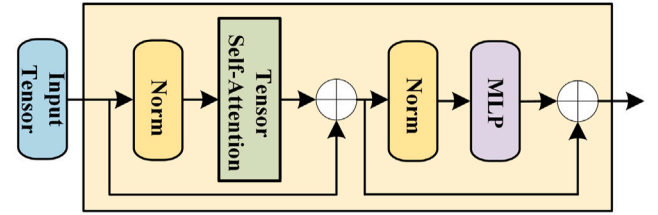


Fig. 2. The operation process of a TEL module.

module remain consistent. Moreover, the TEL module can be employed as a versatile neural network coding layer, enabling the construction of multiple Encoder layers simultaneously to collectively accomplish feature extraction tasks. By leveraging the collaboration of multiple TEL modules, the input tensor is encoded, yielding a sequence of feature representations for the original tensor. These feature representations enable the subsequent network to perform classification tasks effectively.

2.3. Network architecture of Tensor Transformer

By incorporating the proposed TEL module and a fully connected neural network-based classification layer, we employ a Tensor Transformer (TT) network to accomplish HSI classification. Following a sequence of TEL module operations and concluding with operations involving the classification layer, which is based on a fully connected neural network, the ultimate classification output can be directly obtained. The network architecture of the TT is shown in Fig. 3.

The input tensor sample $\mathcal{X} = \{x_{i_1 i_2 i_3}\} \in \mathbb{R}^{I_1 \times I_2 \times I_3}$ that corresponds to a certain pixel is formed by collecting all features of its local neighborhood pixels, where I_3 is the number of spectral bands and $I_1 \times I_2$ is the input spatial size. By sampling the input tensor $\mathcal{X}$ with a spectral band length of J , an input tensor group $\mathcal{Y}_1 \dots \mathcal{Y}_N$ containing N tensors can be obtained. Drawing inspiration from the class token mechanism utilized in BERT [36], we introduced a trainable tensor block into the input three-dimensional tensor group. This block merges with the original tensor group $\mathcal{Y}_1 \dots \mathcal{Y}_N$ to create a new three-dimensional tensor group $\mathcal{Y}_1 \dots \mathcal{Y}_N, \mathcal{Y}_{N+1}$. To enhance the spatial-spectral representation, a tensor group $\mathcal{P}_1 \dots \mathcal{P}_N, \mathcal{P}_{N+1}$ with the same dimensions as $\mathcal{Y}_1 \dots \mathcal{Y}_N, \mathcal{Y}_{N+1}$ is created. The positional encoding within $\mathcal{P}_1 \dots \mathcal{P}_N, \mathcal{P}_{N+1}$ is initialized

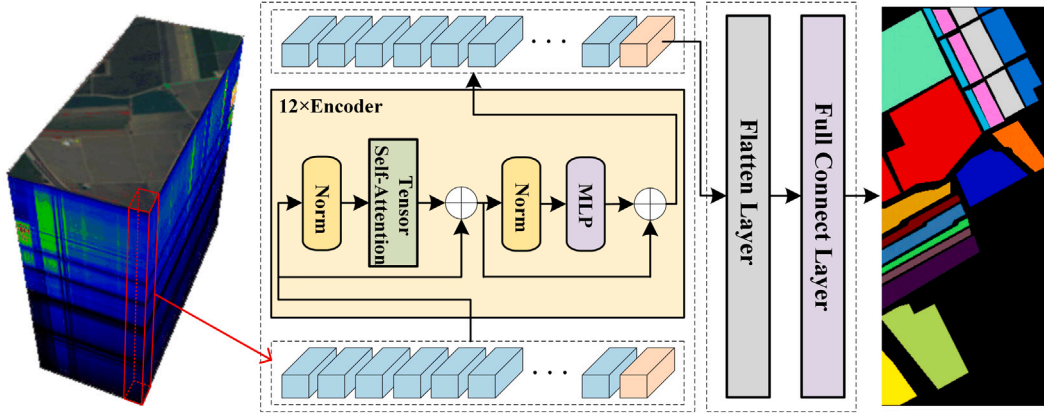


Fig. 3. Network architecture of Tensor Transformer.

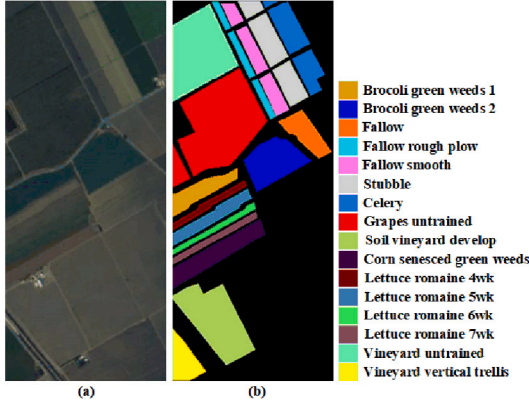


Fig. 4. Images of the SA dataset. (a) RGB image. (b) Ground truth.

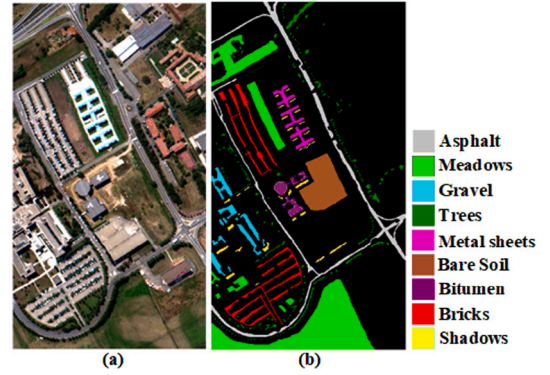


Fig. 5. Images of the PU dataset. (a) RGB image. (b) Ground truth.

using a truncated normal distribution with small variance, which effectively mitigates the risk of gradient vanishing or explosion during the initial training stages. Each positional encoding value is designed to provide unique spatial and spectral positional information, ensuring that the spatial-spectral dependencies are effectively preserved within the input tensor group. Finally, the positional encoding $P_1 \dots P_N, P_{N+1}$ is incorporated into the input tensor group $\mathcal{Y}_1 \dots \mathcal{Y}_N, \mathcal{Y}_{N+1}$ through element-wise addition, embedding the spatial and spectral information directly into the input representation.

Following the aforementioned processing steps, the resultant three-dimensional tensor group is inputted into a 12-layer TEL module. Through 12 iterations of Encoder operations, an output tensor of identical dimensions to the input tensor group is obtained. The output tensor associated with the Class Token tensor is selected as the overall output. Subsequently, the output tensor $\mathcal{Z} = \{z_{i_1 i_2 j}\} \in \mathbb{R}^{I_1 \times I_2 \times J}$ is flattened and fed into the classification layer, which is based on a fully connected neural network, yielding the final classification output of the network.

3. Results and analysis

3.1. Experimental data

(1) Salinas Dataset (SA): The Salinas scene (SA) dataset is a HSI dataset captured by an airborne Visible/Infrared Imaging Spectrometer (AVIRIS) sensor over agricultural land in Salinas, California, United States. The SA dataset comprises 224 spectral bands, covering a wavelength range from 400 to 2500 nm. Each HSI within the dataset has dimensions of 512×217 pixels and a spatial resolution of 3.7 m/pixel.

The dataset encompasses 50,608 labeled pixels, representing 16 different feature types such as fallow and celery. The color image and

corresponding ground truth are shown in Fig. 4. The SA Dataset can be downloaded from https://www.ehu.es/ccwintco/index.php?title=Hyperspectral_Remote_Sensing_Scenes.

(2) Pavia University Scene Dataset (PU): The Pavia University scene (PU) dataset is an HSI collected by a reflection optical system imaging spectrometer (ROSIS) sensor in the urban area of the University of Pavia, Italy. The HSI has dimensions of 610×340 pixels, with a spatial resolution of 1.3 m/pixel. It comprises 103 spectral bands ranging from 430 to 860 nm. The dataset includes a total of 42,776 labeled pixels representing nine feature types, including ash and soil. Fig. 5 displays both the pseudo color and ground truth images of the ground, providing visual representations of the dataset. The PU Dataset can be downloaded from https://www.ehu.es/ccwintco/index.php/Hyperspectral_Remote_Sensing_Scenes#Pavia_Centre_and_University.

(3) WHU-Hi-LongKou Dataset (LK): The WHU-Hi-LongKou dataset (LK) was obtained within the vicinity of Longkou Town, located in the province of Hubei, China. It was captured utilizing an 8-mm focal length Headwall Nano-Hyperspec imaging sensor, which was equipped on a DJI Matrice 600 Pro unmanned aerial vehicle (UAV) platform. The study area encompasses a simplistic agricultural landscape, housing six distinct crop species: corn, cotton, sesame, broad-leaf soybean, narrow-leaf soybean, and rice. The UAV conducted its survey at an altitude of 500 m, generating imagery with dimensions of 550×400 pixels. The dataset comprises 270 spectral bands spanning from 400 to 1000 nm, and the spatial resolution of the HSI acquired by the UAV is approximately 0.463 m. An overview of this dataset can be found in Fig. 6. The LK dataset can be accessed from http://rsidea.whu.edu.cn/resource_WHUHi_sharing.htm.

(4) Houston2013 Dataset (Hou): The Houston2013 dataset (Hou) was captured using the ITRES CASI-1500 sensor over the University

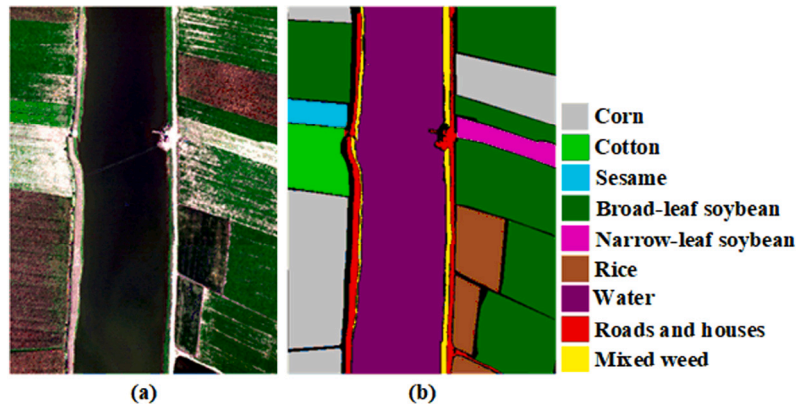


Fig. 6. Images of the LK dataset. (a) RGB image. (b) Ground truth.

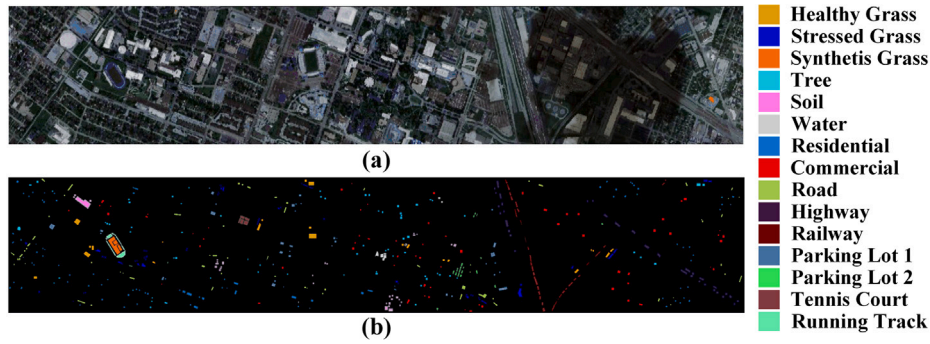


Fig. 7. Images of the Hou dataset. (a) RGB image. (b) Ground truth.

of Houston and its neighboring rural areas in Texas, USA. The Houston2013 dataset includes 144 spectral bands with a wavelength range from 380 to 2510 nm. The dimensions of the image are 349×1905 pixels, with a spatial resolution of 2.5 meters per pixel. This dataset encompasses 15 challenging land cover and land use categories, as detailed in Table 4. The visualization of the false-color HSI and the released training and testing samples from the 2013 IEEE GRSS Data Fusion Contest are presented in Fig. 7. The Hou dataset can be accessed from https://hyperspectral.ee.uh.edu/?page_id=459.

3.2. Experimental results

(1) Experimental Setup: The datasets used in our experiments are all standard versions provided by official sources. Before starting the experiments, we applied the same preprocessing steps to each dataset. First, we performed per-band standardization to eliminate differences in feature scales. Then, we conducted global normalization to further smooth the data distribution across the entire dataset.

All competitors were trained and implemented on our computing server equipped with two Intel Xeon E5-2620 CPUs and four Nvidia TITAN XP GPUs. During the training phase, a batch size of 256 was set, an initial learning rate of 0.001 was used, and model optimization was performed using the Adam [37] optimizer. For backpropagation, the cross-entropy loss function was employed. To quantitatively assess the performance of the proposed method, we utilized metrics such as overall classification accuracy (OA), average accuracy (AA), and kappa coefficient (Kappa).

To verify the effectiveness of the proposed method for HSI classification, we performed experiments on the SA, PU, LK and Hou datasets. We compared the TT method with other existing approaches, including the classical PCA-SVM [11] method, the state-of-the-art SAE-CNN [14], RSSAN [15], A²S²K-ResNet [26], SpectralFormer [29], HiT [30], morphFormer [31] and SSFTT [32] methods.

To maintain consistency with the original paper, the dimensions of the input data were reduced to 30 using PCA for the SSFTT method, while for SAE-CNN, the data was dimensionally reduced to 5 using SAE. However, it is important to note that SAE-CNN, RSSAN, A²S²K-ResNet, SpectralFormer, HiT, morphFormer and the proposed TT approaches are single end-to-end models for HSI classification and do not require specialized dimensionality compression models. For the TT, 3D CNN, RSSAN, ViT, SpectralFormer, HiT and SSFTT classifiers, the input spatial sizes were set to 11×11 for the SA dataset, 9×9 for the PU dataset and 7×7 for the LK and Hou dataset.

For SVM classifiers, the input consists of individual samples without considering neighboring pixels. In our experiment, we randomly selected labeled pixels per class for training data in the SA, PU, and LK datasets. For the Hou dataset, we used the standard training set provided by the official partitioning for training. The training epoch was set to 300. For the input sample selection of TT, a spectral length of 20 with a stride of 10 was used for the SA dataset; for PU, a spectral length of 25 with a stride of 5; for LK, a spectral length of 54 with a stride of 27; and for Hou, a spectral length of 54 with a stride of 30. All tensors were sampled with overlapping regions. What is more, the TT method employed 12 TEL modules. Each experiment was repeated 10 times with randomly selected training samples to ensure statistical robustness.

(2) Classification Result of Different Methods: To evaluate the effectiveness of the proposed method for HSI classification, we conducted extensive and comprehensive comparative experiments on the SA, PU, LK, and Hou datasets. The SA and PU datasets have a sufficient number of samples, and we randomly selected 100 labeled pixels per class for the training data. The classification results of SAE-CNN, RSSAN, A²S²K-ResNet, SpectralFormer, HiT, morphFormer, SSFTT and the TT methods are presented in Figs. 8–9. By visualizing the classification results, we observed that TT achieves the closest results to the ground truth on both datasets. A quantitative comparison of the classification accuracy are

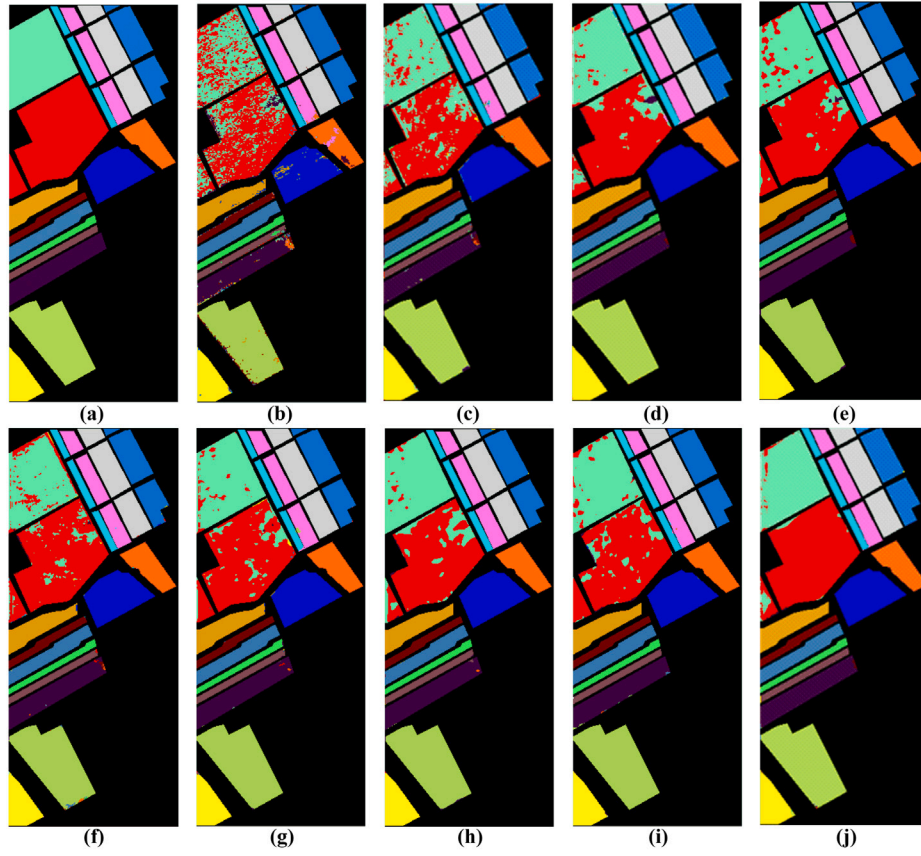


Fig. 8. Classification result of different methods using SA dataset. (a) Groundtruth, (b) PCA-SVM, (c) SAE-CNN, (d) RSSAN, (e) A^2S^2K -ResNet, (f) SpectralFormer, (g) HiT, (h) morphFormer, (i) SSFTT, (j) TT.

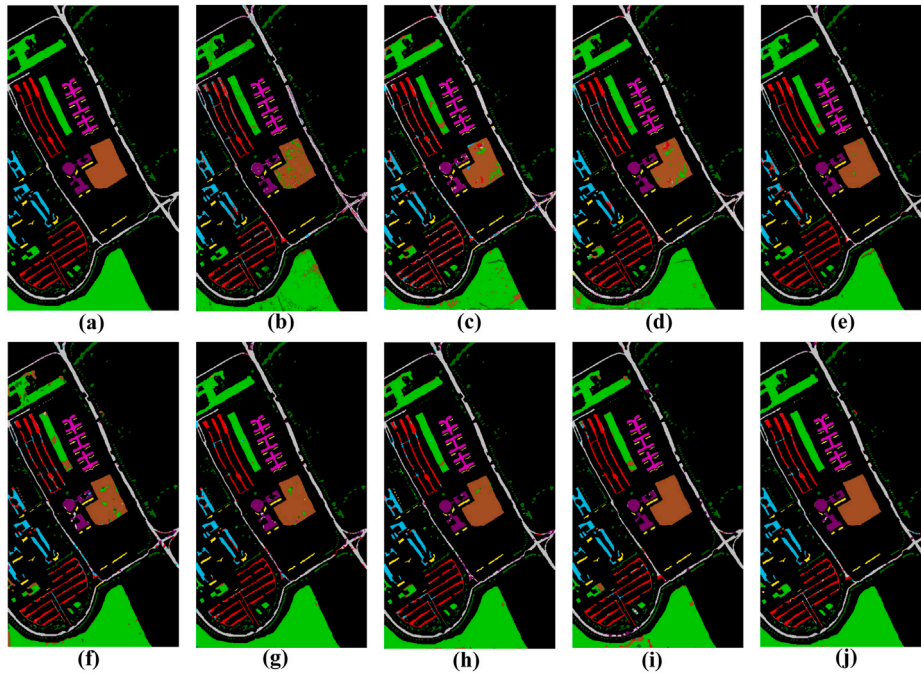


Fig. 9. Classification result of different methods using PU dataset. (a) Groundtruth, (b) PCA-SVM, (c) SAE-CNN, (d) RSSAN, (e) A^2S^2K -ResNet, (f) SpectralFormer, (g) HiT, (h) morphFormer, (i) SSFTT, (j) TT.

Table 1

Classification performances of different methods on SA dataset.

Class	Train sample	PCA-SVM [11]	SAE-CNN [14]	RSSAN [25]	A ² S ² K-ResNet [26]	SF [29]	HiT [30]	Morph-Former [31]	SSFTT [32]	TT
1	100	96.81	99.75	98.31	99.95	99.20	99.70	100.0	100.0	99.94
2	100	96.32	99.54	100.0	100.0	99.97	99.84	100.0	99.95	99.87
3	100	61.89	99.81	99.88	100.0	99.85	100.0	100.0	100.0	99.90
4	100	98.85	99.78	100.0	99.86	99.64	98.28	100.0	99.64	98.83
5	100	97.20	99.80	99.96	99.85	99.89	99.03	99.89	99.03	99.81
6	100	98.08	99.95	100.0	100.0	99.90	100.0	100.0	100.0	100.0
7	100	99.30	99.88	99.69	99.78	99.27	99.92	99.78	99.94	99.77
8	100	97.45	84.13	88.32	90.01	90.42	88.17	85.25	88.03	97.67
9	100	99.03	99.44	99.64	99.23	98.74	99.89	99.47	100.0	99.77
10	100	71.48	97.46	100.0	98.38	97.99	99.00	98.96	98.51	100.0
11	100	71.20	99.51	99.47	99.72	99.81	99.81	100.0	100.0	99.61
12	100	80.28	99.91	99.93	100.0	99.90	100.0	99.58	99.95	99.95
13	100	65.98	99.98	100.0	100.0	100.0	99.02	100.0	100.0	100.0
14	100	88.22	99.20	100.0	99.91	98.88	99.25	99.53	99.72	100.0
15	100	81.05	87.02	96.56	88.30	86.03	93.70	96.20	93.45	98.19
16	100	59.60	99.76	99.70	100.0	99.61	99.34	99.72	99.39	98.55
OA (%)		87.44	94.62	97.04	96.13	95.71	96.44	96.24	96.47	99.13
AA (%)		85.23	97.81	98.94	98.44	98.07	98.43	98.65	98.60	99.49
Kappa (%)		86.20	94.02	96.71	95.70	95.23	96.04	95.82	96.05	99.03
Para		–	0.1M	0.1M	0.4M	6.4M	24.7M	0.3M	0.2M	0.1M
FLOPs		–	5.1M	8.0M	175.7M	362.6M	707.1M	25.6M	4.2M	91.3M
PreTime		–	11.0s	23.1s	33.9s	35.0s	43.0s	39.9s	20.1s	29.6s

Table 2

Classification performances of different methods on PU dataset.

Class	Train sample	PCA-SVM [11]	SAE-CNN [14]	RSSAN [25]	A ² S ² K-ResNet [26]	SF [29]	HiT [30]	Morph-Former [31]	SSFTT [32]	TT
1	100	91.28	90.17	97.48	95.37	90.18	90.02	91.80	90.45	92.78
2	100	91.49	92.63	88.18	96.03	93.75	98.55	97.22	94.03	99.39
3	100	41.21	89.30	94.95	75.61	90.32	88.80	95.47	96.86	94.54
4	100	66.68	98.78	93.15	99.35	94.78	99.48	97.19	98.86	98.33
5	100	98.96	99.99	100.0	100.0	99.99	99.93	100.0	100.0	100.0
6	100	57.04	93.19	99.56	99.07	98.31	97.40	99.07	99.86	99.94
7	100	65.47	95.38	95.86	97.74	94.74	94.59	98.27	98.57	96.47
8	100	94.13	91.40	97.64	97.34	93.54	95.87	87.99	90.52	89.16
9	100	99.68	99.77	83.00	100.0	98.79	100.0	99.79	99.89	99.89
OA (%)		82.99	90.54	92.96	95.90	92.66	96.40	95.89	94.80	97.12
AA (%)		78.44	91.49	94.42	95.61	93.05	96.07	96.31	96.56	96.72
Kappa (%)		79.03	90.13	90.86	94.61	90.64	95.25	94.59	93.20	96.25
Para		–	0.1M	94.6K	0.2M	5.2M	13.5M	0.2M	0.2M	0.3M
FLOPs		–	3.5M	5.6M	88.2M	321.0M	285.5M	15.6M	4.2M	110.2M
PreTime		–	10.6s	15.6s	19.2s	21.4s	18.5s	35.9s	16.2s	19.3s

provided in Tables 1–2. It is noteworthy that the proposed TT method achieves the best performance in terms of the OA, AA and the Kappa coefficient among all the methods. In terms of OA, the TT method outperforms SSFTT by 2.66%, morphFormer by 2.89%, HiT by 2.69%, SpectralFormer by 3.42%, A²S²K-ResNet by 3.00%, RSSAN by 2.09%, SAE-CNN by 4.51%, and PCA-SVM by 11.69%, based on the classification results on the SA dataset. When considering the PU dataset, the TT method improves 2.32%, 1.23% 0.72%, 4.46%, 1.22%, 6.58% and 14.13% compared to SSFTT, morphFormer, HiT, SpectralFormer, A2S2K-ResNet, RSSAN, SAE-CNN and PCA-SVM, respectively.

Based on the information provided in Tables 1–2, it is evident that the SVM classifier exhibits lower classification accuracy compared to the other classifiers. This discrepancy can be attributed to the relatively simplistic structure and fewer parameters of the SVM model. Consequently, the SVM classifier struggles to achieve highly accurate classification results, particularly in complex classification scenarios where more sophisticated models may be better equipped to handle the intricacies of the data. Regarding the SAE-CNN method, it is important to note that there exists a hard coupling between the feature compression model and the classification model. This coupling means that the effectiveness of the input features for the classifier is not guaranteed after dimensionality reduction by the data compression model. While these classifiers maintain an end-to-end structure, the limitations of the feature compression model can potentially lead to less discriminative input features for the classifier. As a result, the classification accuracy may be impacted. Although RSSAN achieves favorable results

by extracting spatial and spectral features through its spectral attention module and spatial attention module, it is worth noting that the two-stage extraction approach employed by RSSAN may disrupt the inherent spatial-spectral fusion structure present in the original data. This disruption may lead to a bottleneck in terms of classification performance. A²S²K-ResNet achieve the competitive performance, suggesting that it has achieved near-optimal results within the CNN architecture. However, the bottleneck may arise from its inability to capture long-range dependencies in the spectral dimension. Even though vision transformers demonstrate capability in capturing extensive spatial and spectral features, the transformer-based approach SpectralFormer still yield unsatisfactory performance. This is because that SpectralFormer disregard the 3-D characteristics of HSI data, thereby fails to adequately extract the spatial-spectral structural features of HSI data. Additionally, HiT, MorphFormer, and SSFTT all achieved good results on the SA dataset, which is benefited from their excellent backbone designs. However, their classification results for the eighth class, Grapes untrained, were relatively poor, with none surpassing 90%. For SSFTT, this may be due to the fact that it simplifies the semantic representation of features using feature tokens, and the tokenization process alone is not sufficient to capture all the essential spectral and spatial details. It is noteworthy to mention that the notable efficacy of the TT approach can be ascribed to its end-to-end network architecture. The core TSAM module of the TT approach enables the concurrent extraction of abundant structural characteristics from both spatial and spectral domains, resulting in a significant enhancement in classification precision.

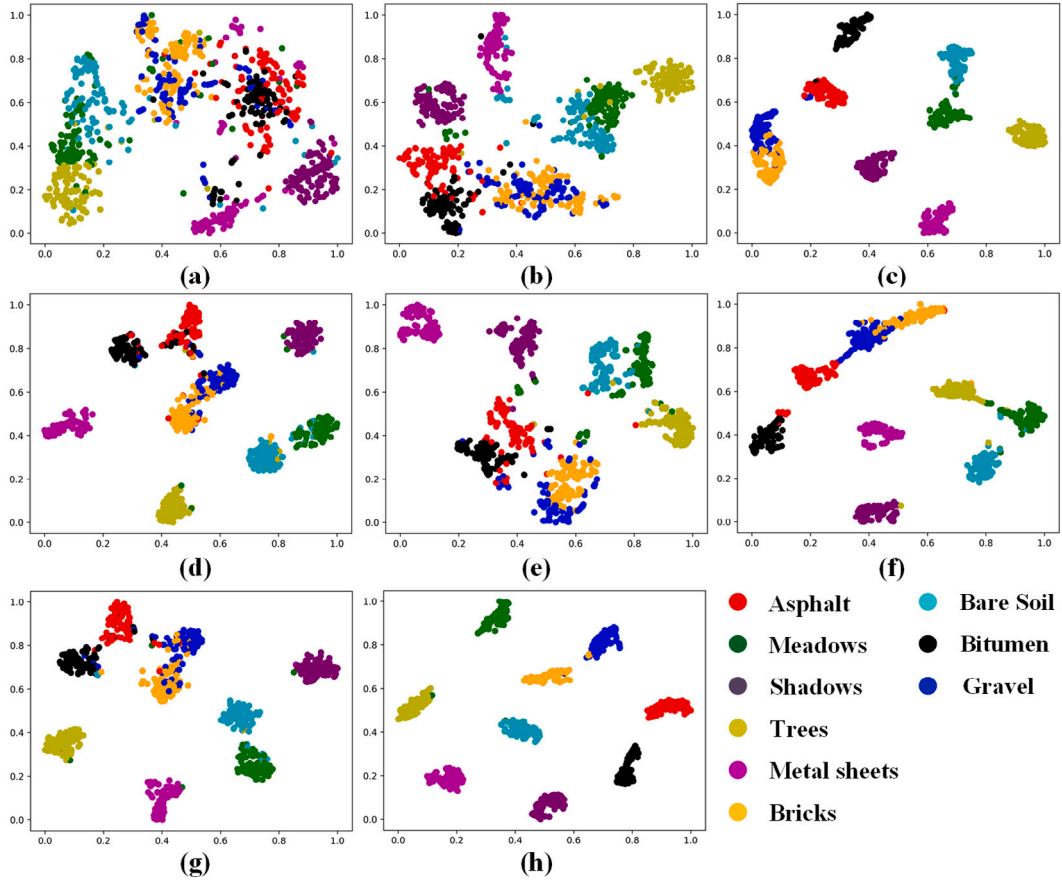


Fig. 10. t-SNE visualization after feature extraction of different methods using PU dataset. (a) SAE-CNN, (b) RSSAN, (c) A^2S^2K -ResNet, (d) SpectralFormer, (e) HiT, (f) morphFormer, (g) SSFTT, (h) TT.

In our previous study [38], we discovered that the spectral signatures of the eighth and fifteenth categories exhibit considerable resemblance, thereby posing substantial difficulties in both feature extraction and classification. It can be concluded from Table 1 that the TT approach achieves superior classification accuracy in the eighth and fifteenth categories, where other methods struggled to differentiate similar spectra and failed to reach 90% accuracy in both categories. The visualization results in Fig. 8 further illustrate this point. This clearly demonstrates the exceptional discriminatory capability of the TT method in discerning these intricately confusable categories.

To further validate the TT model's ability to extract spatial and spectral features from the original data, we used t-distributed stochastic neighbor embedding (t-SNE) to visualize and analyze the feature mappings obtained by different methods. We compared the TT model with SAE-CNN, RSSAN, A^2S^2K -ResNet, SpectralFormer, HiT, morphFormer and SSFTT. The visualization results are shown in Fig. 10. We see that SAE-CNN, RSSAN methods are unable to effectively distinguish the features from different classes, resulting in lower classification accuracy. A^2S^2K -ResNet, morphFormer, and TT methods achieved the best feature clustering performance. However, A^2S^2K -ResNet and morphFormer struggled to clearly separate the classes "Bricks" and "Gravel." In contrast, the TT method successfully distinguished all categories, while exhibiting the largest inter-class variance and the smallest intra-class variance in feature clustering.

To investigate the classification performance of the TT method under small-sample conditions, we selected a smaller number of training samples on the LK dataset. The experimental visualization results are shown in Fig. 11, and Table 3 presents a quantitative comparison of the classification accuracy. It is evident that, compared to the experimental results on the SA and PU datasets, the classification accuracy

of transformer-based methods decreases on the LK dataset. This is because transformer models require larger amounts of data, as their self-attention mechanism relies on learning global relationships among features from large datasets. When the sample size is insufficient, these relationships may not be adequately captured, thereby affecting model performance. In contrast to transformer-based methods, SAE-CNN, RSSAN, and A^2S^2K -ResNet achieved results closer to expectations, with A^2S^2K -ResNet yielding the best performance among all methods. However, the TT method also performed competitively, with its OA being comparable to A^2S^2K -ResNet and its Kappa slightly higher. A^2S^2K -ResNet is well-known for its strong classification performance under small-sample conditions, and the fact that the TT method, as a transformer-based approach, can achieve nearly equivalent results demonstrates its superior performance over other transformer-based methods under small-sample scenarios.

What is more, Tables 1–4 lists the number of parameters (Para), FLOPs, and prediction time (PreTime) for all methods except SVM. Based on the experimental results across four datasets, the number of parameters in TT is comparable to methods based on CNN and attention mechanisms, and significantly lower than other transformer-based methods except for SSFTT and morphFormer. The TT method has a simpler network structure and fewer parameters while still achieving higher classification accuracy. Among transformer-based methods, SSFTT has less parameter count. This is due to the dimensionality reduction of the raw data via PCA in advance. However, it disrupts the spatial-spectral features of the HSI. In contrast, TT method takes multiple tensor blocks obtained from spectral segmentation as the input, which retains the spatial-spectral feature structure of the original data, resulting in higher overall classification accuracy. Unfortunately, the FLOPs and PreTime are higher than that of SSFTT because of the large-scale input samples.

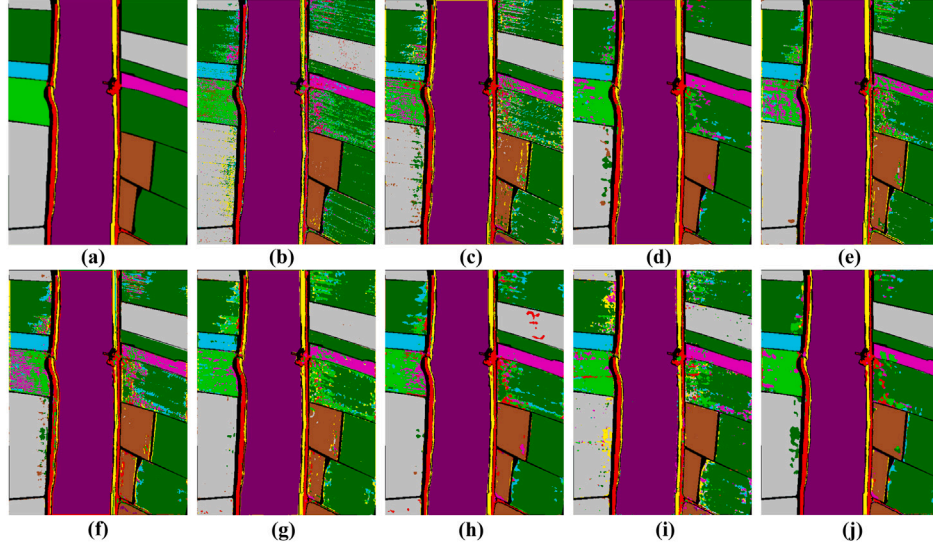


Fig. 11. Classification result of different methods using LK dataset. (a) Groundtruth, (b) PCA-SVM, (c) SAE-CNN, (d) RSSAN, (e) A^2S^2K -ResNet, (f) SpectralFormer, (g) HiT, (h) morphFormer, (i) SSFTT, (j) TT.

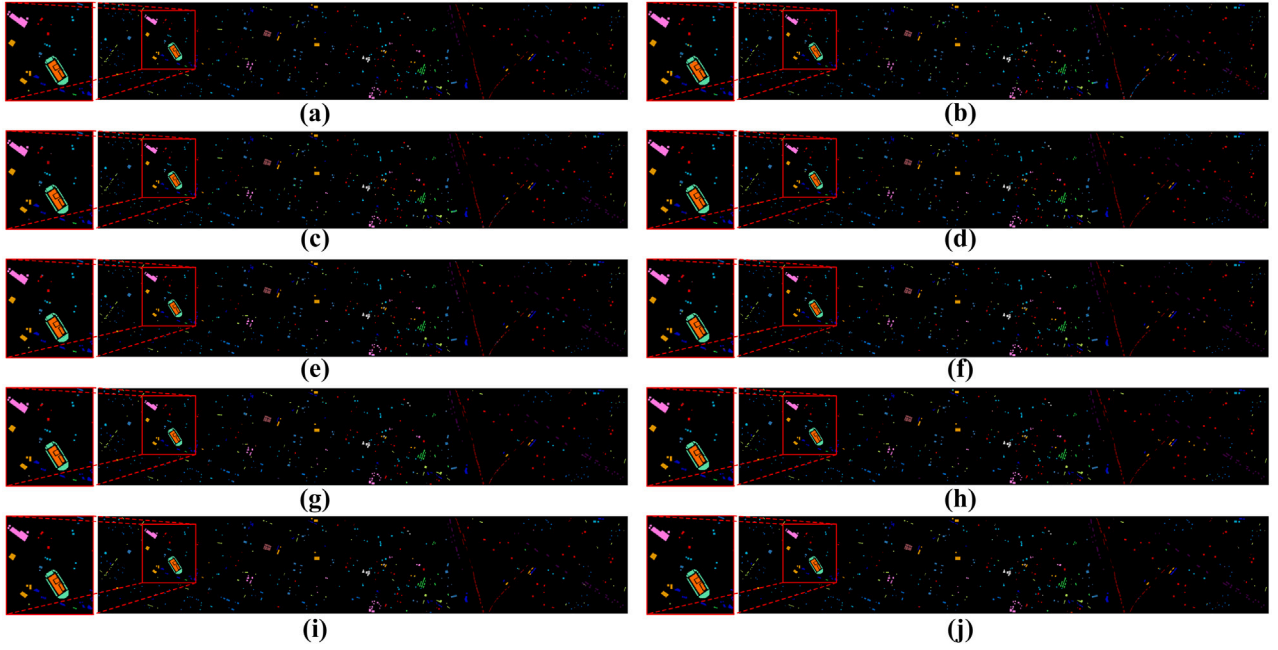


Fig. 12. Classification result of different methods using Hou dataset. (a) Groundtruth, (b) PCA-SVM, (c) SAE-CNN, (d) RSSAN, (e) A^2S^2K -ResNet, (f) SpectralFormer, (g) HiT, (h) morphFormer, (i) SSFTT, (j) TT.

Finally, we conducted comparative experiments on the Hou dataset. Notably, we used the officially partitioned training set, ensuring that the evaluation of our method avoids the potential overlap issue between training and testing patches, providing a more accurate reflection of the method's performance [39]. The classification results for the Hou dataset are presented in Fig. 12 and Table 4. Compared to other transformer-based methods, both SSFTT and TT exhibit significantly fewer parameters and less PreTime while still achieving higher classification accuracy. Specifically, in terms of OA, the TT method shows improvements of 0.33%, 1.61%, 1.39%, 2.37%, 1.33%, 3.16%, 4.13%, and 17.17% compared to SSFTT, morphFormer, SpectralFormer, A^2S^2K -ResNet, RSSAN, SAE-CNN, and PCA-SVM, respectively.

4. Discussion

4.1. Discussion on different learning rates

The learning rate holds utmost importance as a hyperparameter in neural network training, as it dictates the pace at which model parameters are updated. A high learning rate can induce excessive fluctuations in parameter values during training, impeding convergence towards the optimal solution. This can result in insufficient model performance or complete failure to converge. Conversely, a low learning rate can decelerate the convergence of the model, necessitating an extended training duration. It may also cause the model to oscillate around local optima, impeding convergence towards the global optimum.

Table 3

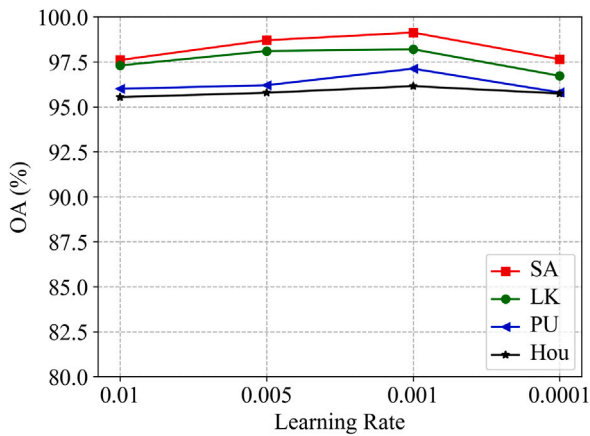
Classification performances of different methods on LK dataset.

Class	Train sample	PCA-SVM [11]	SAE-CNN [14]	RSSAN [25]	A ² S ² K-ResNet [26]	SF [29]	HiT [30]	Morph-Former [31]	SSFTT [32]	TT
1	20	86.78	93.66	98.77	95.88	96.26	98.63	97.47	94.98	96.04
2	20	69.49	63.05	57.93	85.67	54.85	83.21	75.22	83.54	87.09
3	20	81.01	83.60	95.25	97.79	93.73	89.67	98.42	98.28	99.37
4	20	72.60	76.56	84.45	92.43	81.21	79.52	85.27	80.56	92.00
5	20	82.69	70.46	66.83	93.71	71.43	75.64	96.34	94.17	86.05
6	20	93.89	84.23	91.77	98.38	91.18	91.45	94.91	97.18	94.05
7	20	94.88	98.61	98.94	98.85	97.24	98.40	99.64	99.00	99.94
8	20	75.33	92.15	82.69	93.50	85.44	92.15	92.32	90.16	95.94
9	20	55.56	80.82	95.51	98.76	83.63	91.30	96.73	96.25	95.64
OA (%)		83.26	87.20	90.98	95.45	88.70	90.59	93.15	91.40	95.42
AA (%)		79.93	82.57	85.79	94.89	88.89	88.89	92.93	92.68	94.01
Kappa (%)		78.58	83.55	88.36	93.84	85.50	87.92	91.15	88.94	94.03
Para		–	0.1M	0.2M	0.5M	7.6M	16.3M	0.3M	0.2M	0.2M
FLOPs		–	3.6M	5.8M	141.0M	359.1M	623.8M	20.8M	4.2M	54.1M
PreTime		–	58.5s	82.8s	144.9s	182.8s	605.6s	184.7s	96.3s	148.3s

Table 4

Classification performances of different methods on Hou dataset.

Class	Train sample	PCA-SVM [11]	SAE-CNN [14]	RSSAN [25]	A ² S ² K-ResNet [26]	SF [29]	HiT [30]	Morph-Former [31]	SSFTT [32]	TT
1	198	62.26	98.96	94.48	99.76	96.64	95.36	96.32	98.24	98.96
2	190	77.02	97.69	99.68	96.89	98.09	99.76	97.85	97.85	99.76
3	192	67.79	99.43	99.57	99.86	99.71	99.71	99.86	100.0	99.71
4	188	76.03	97.59	88.18	98.38	88.91	97.27	94.37	98.31	98.79
5	186	71.38	99.44	97.34	99.84	99.84	99.44	99.60	100.0	99.92
6	182	81.38	99.69	96.62	100.0	99.69	99.38	100.0	100.0	100.0
7	196	81.51	91.01	92.27	95.19	93.14	94.16	90.69	96.06	97.16
8	191	88.84	92.52	95.18	86.66	90.59	94.45	95.42	93.65	94.13
9	193	86.93	86.50	86.10	95.37	93.69	93.45	95.77	96.88	95.29
10	191	60.02	99.19	98.70	99.02	98.21	99.27	99.59	98.21	99.19
11	181	77.78	88.99	95.71	98.22	98.85	98.38	99.60	98.22	100.0
12	192	63.57	81.43	98.62	96.76	97.73	96.11	94.40	100.0	97.59
13	184	87.81	95.74	95.10	94.46	92.32	92.54	95.74	100.0	99.57
14	181	96.45	100.0	100.0	100.0	100.0	100.0	100.0	100.0	100.0
15	187	96.21	100.0	99.70	100.0	100.0	99.85	100.0	99.70	100.0
OA (%)		81.27	94.31	95.28	97.11	96.07	97.05	96.83	98.11	98.44
AA (%)		78.33	95.21	95.82	97.44	96.50	97.28	97.28	98.47	98.72
Kappa (%)		80.72	93.86	94.89	96.87	95.75	96.81	96.58	97.96	98.32
Para		–	0.1M	0.1M	0.3M	3.6M	7.7M	0.2M	0.2M	0.1M
FLOPs		–	11.5M	4.0M	74.3M	155.3M	257.2M	11.6M	18.5M	22.3M
PreTime		–	7.0s	7.6s	10.6s	15.2s	15.7s	14.8s	7.3s	9.2s

**Fig. 13.** Classification accuracy of the TT method trained with different learning rates.

To determine the optimal learning rate for the TT model, a series of experiments were conducted on the aforementioned four datasets. It is evident from Fig. 13 that as the learning rate decreases, the classification accuracy of the TT model on the SA, PU, LK and Hou datasets exhibits an initial increase followed by a subsequent decrease. The highest classification performance of the TT model is evident at a learning rate of 0.001. This observation suggests that as the learning

rate decreases, the model undergoes more comprehensive training, leading to enhanced classification accuracy. Nevertheless, it is worth noting that when the learning rate decreases to 0.0001, the model's performance may decline due to an excessively low learning rate. In such cases, the model may struggle to converge effectively within the designated training period. Hence, in order to achieve a balance between model accuracy and training cost, a learning rate of 0.001 will be utilized for subsequent experiments.

4.2. Discussion on different input spectral size

Overlapping sampling is usually adopted to segment the spectrum of HSI data. If the segmented spectral length is short and the sampling stride is small, which lead to increased tensor blocks for input and higher computational complexity. Conversely, if the segmented spectral length is long and sampling stride is too large, the input tensor blocks is less, TT model may fail to capture subtle spectral differences, resulting in poor classification accuracy. Therefore, finding an appropriate balance between sampling parameters is crucial.

Using the Hou Dataset as an example, we performed experiments with various combinations of the sampling stride and spectral length. The results are shown in Fig. 14, we see that the model achieves the best classification performance when spectral length is 54 and the sampling stride is 30, with the OA of 96.15%, AA of 95.91%, and Kappa of 0.957. For other dataset, one can also obtain corresponding hyper parameters.

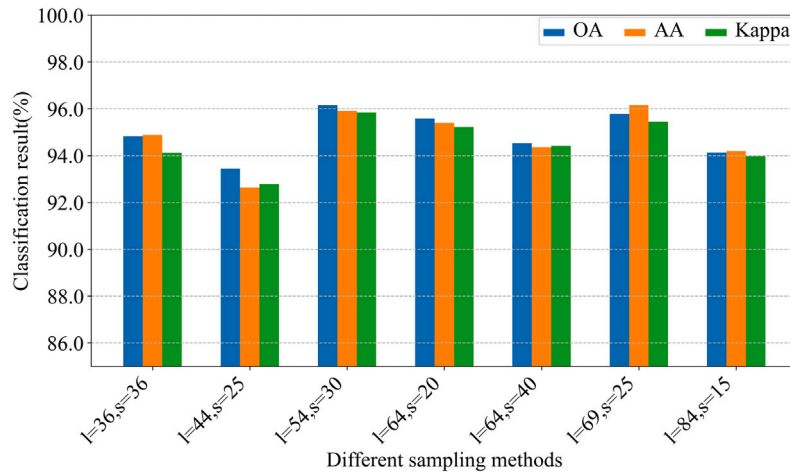


Fig. 14. Classification accuracy of different input spectral sizes.

Table 5

Performance and computational costs of the TT method with varying input spatial sizes and Encoder layers on PU dataset.

Encoder layers	Win size											
	5 × 5			7 × 7			9 × 9			11 × 11		
	OA(%)	Para	Flops	OA(%)	Para	Flops	OA(%)	Para	Flops	OA(%)	Para	Flops
1	93.87	37.7K	2.6M	94.02	60.6K	5.3M	94.37	91.0k	9.2M	93.49	0.1M	14.5M
4	94.77	80.1K	10.2M	95.83	0.1M	21.1M	96.52	0.1M	36.7M	93.95	0.2M	57.6M
12	95.12	0.2M	30.6M	96.71	0.2M	63.3M	97.12	0.3M	110.2M	94.31	0.3M	172.7M
16	95.01	0.2M	40.8M	96.59	0.3M	84.4M	96.98	0.3M	146.9M	94.29	0.4M	230.3M

4.3. Discussion on different number of input spatial size and tensor encoder layers

For transformer-based HSI classification models, input spatial size and encoder layers are two critical hyperparameters. A small input spatial size may omit important spatial details, while an excessively large size risks overfitting to noise, compromising generalization. Similarly, more encoder layers improve feature extraction but increase computational costs and overfitting risks. Therefore, finding a balance between Input Spatial Size and encoder layers is essential for achieving optimal classification performance and computational efficiency in transformer-based HSI classification models.

To investigate the impact of input spatial size and the number of encoder layers on the TT method, we conducted a series of experiments using the PU dataset as an example. 100 samples were selected from each class, and all other parameter settings were kept consistent with previous experiments. The results are summarized in Table 5, where the horizontal axis represents the input spatial size, and the vertical axis represents the number of encoder layers.

For the input spatial size, it is evident that smaller sizes, such as 5×5 , result in suboptimal OA across all encoder layer configurations, with the highest OA being 95.12% when using 12 encoder layers. This can be attributed to the limited spatial context captured by smaller windows, which hinders the model's ability to fully utilize spatial information. Conversely, larger input spatial sizes, such as 11×11 , lead to increased computational costs in terms of both parameters and FLOPs, reaching as high as 0.4M parameters and 230.3M FLOPs under the 16-layer configuration. However, despite the greater computational demands, the performance does not improve significantly compared to the 9×9 spatial size. The 9×9 spatial size consistently outperforms other configurations, achieving the highest OA of 97.12% with 12 encoder layers, while maintaining a moderate computational cost of 0.3M parameters and 110.2M FLOPs. Regarding the number of encoder layers, increasing the layers enhances the model's capacity to capture complex features, as reflected in the improved OA from 94.37% to 97.12% when transitioning from 1 to 12 layers for the 9×9 spatial size. However,

further increasing the number of layers to 16 does not yield substantial accuracy improvements and instead results in a significant rise in computational costs. This indicates diminishing returns in performance beyond 12 layers, where the risk of overfitting and computational overhead outweigh the benefits of deeper representations. Taken together, these findings highlight the 9×9 spatial size and 12 encoder layers as the optimal configuration for balancing classification performance and computational efficiency in the TT method.

4.4. Ablation study

To validate the effectiveness of the TSAM in extracting long-range spatial-spectral structural features from the original data and enhancing classification performance, a series of experiments were conducted. The TSAM module was substituted with the spatial correlation module and the spectral correlation module individually.

The first approach, known as the Spatial Transformer, solely focuses on spatial attention. In this approach, the TSAM module in the TT model is replaced with a conventional Self-Attention mechanism. The input data, with spatial sizes of 11×11 for the SA dataset, 9×9 for the PU dataset, 7×7 for the LK and Hou datasets, is partitioned into individual pixels and served as the input for the Spatial Transformer. Similarly, the second approach, called the Spectral Transformer, exclusively exploits spectral attention. Still using the traditional self-attention mechanism instead of the TSAM module. The input sequence for the Spectral Transformer is divided into multiple sequences based on specific rules, where each sequence contains a subset of spectral information for a particular pixel. The sampling methods for input samples in the TT method remain consistent with Section 3. All other experimental parameters are maintained in accordance with the preceding description. The experimental results for the Spatial Transformer, Spectral Transformer, and the TT method are presented in Table 6. The results presented in Table 6 illustrate the classification accuracy of the Spatial Transformer, Spectral Transformer, and TT methods across four datasets. It is evident that the TT method significantly outperforms both the Spatial Transformer and Spectral Transformer

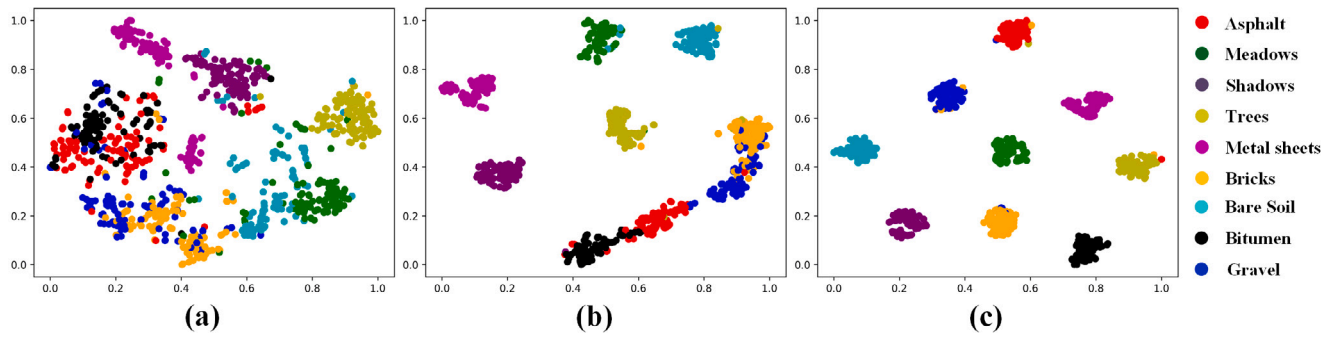


Fig. 15. t-SNE visualization of feature representations on the PU dataset. (a) Spectral Transformer, (b) Spatial Transformer, (c) TT.

Table 6

Comparison of classification accuracy of different methods.

Methods	SA	PU	LK	Hou
Spatial Transformer	96.82	93.76	96.27	94.43
Spectral Transformer	95.34	91.25	93.42	93.71
TT	99.13	97.12	98.20	96.15

across all datasets. Specifically, the TT method achieves the highest OA on the SA dataset (99.13%), PU dataset (97.12%), LK dataset (98.20%), and Hou dataset (96.15%). This performance improvement highlights the effectiveness of the TSAM in jointly leveraging spatial and spectral attention to extract long-range spatial-spectral structural features from raw hyperspectral image data. In contrast, the Spatial Transformer and Spectral Transformer, which focus on spatial and spectral attention individually, show relatively lower classification accuracies. The t-SNE visualization in Fig. 15 further supports the analysis of the classification performance. From the visualizations, it is evident that the TT method achieves significantly better clustering performance compared to the other two approaches. In subfigure (c), the data points for each class are tightly grouped, with clear boundaries between clusters, demonstrating the TT method's ability to effectively extract and utilize both spatial and spectral features for classification. In contrast, subfigure (a) shows that the Spectral Transformer results in overlapping and scattered clusters, indicating weaker class separability. Similarly, in subfigure (b), while the Spatial Transformer performs slightly better than the Spectral Transformer, it still exhibits overlapping clusters and less distinct class boundaries. These results highlight the superior capability of the TT method in leveraging the intrinsic spatial-spectral structure of the hyperspectral data, leading to more accurate classification.

5. Conclusion

In this paper, we propose a novel network model called TT for the HSI classification task. TT is an end-to-end network that directly takes in raw 3-D HSI data as the network input, without requiring prior dimensionality reduction. The fundamental constituent, namely the Tensor Self-Attention Mechanism (TSAM) and Tensor Encoder Layer (TEL), empower the network to concentrate on extracting long-range spatial-spectral structural features within raw 3-D HSI data. This endeavor ultimately leads to an enhanced outcome in HSI classification.

To evaluate the effectiveness of the proposed method, extensive experiments were conducted on several benchmark datasets, including SA, PU, LK, and Hou. The TT model demonstrated excellent classification results across all datasets, consistently outperforming other state-of-the-art methods. This highlights the robustness and effectiveness of the TT model in handling diverse hyperspectral image classification tasks. However, certain limitations exist in our current work. A primary challenge is the selection of input spectral size. While the TT model performs well with the chosen spectral size, further experimentation is needed to determine the optimal size for different datasets. We aim

to develop an adaptive input spectral size module for the TT network to improve scalability and handle diverse HSI data characteristics. Additionally, while the TT model achieves high classification accuracy, its computational cost, in terms of FLOPs and prediction time, is higher compared to other transformer-based methods like SSFTT. This is due to the large-scale input samples that preserve the spatial-spectral feature structure of the original data but require more resources. Future work will focus on optimizing the model's parameter count and processing efficiency to improve scalability and reduce computational costs.

CRedit authorship contribution statement

Wei-Tao Zhang: Writing – original draft. **Yv Bai:** Writing – original draft. **Sheng-Di Zheng:** Writing – review & editing. **Jian Cui:** Methodology. **Zhen-zhen Huang:** Visualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

The data that has been used is confidential.

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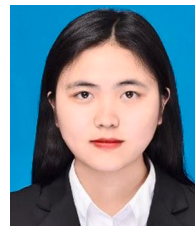
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